

LECTURE 5: SWITCH RELUCTANCE MOTOR DRIVES

The SRM drive is considered as an attractive candidate for variable speed motor drives due to its low cost, rugged structure, reliable converter topology, high efficiency over a wide speed range, and simplicity in control. These drives are suitable for EVs, HEV traction applications, aircraft starter/generator systems, mining drives, washing machines, door actuators, and so on.

The SRM has a simple, rugged, and low-cost structure. It has no PM or winding on the rotor. This structure not only reduces the cost of the SRM, but also offers high-speed operation capability for this motor. Unlike the induction and PM machines, the SRM is capable of high-speed operation without the concern of mechanical failures that result from the high-level centrifugal force. In addition, the inverter of the SRM drive has a reliable topology. The stator windings are connected in series with the upper and lower switches of the inverter. This topology can prevent the shoot-through fault that exists in the induction and permanent motor drive inverter. Moreover, high efficiency over a wide speed range and control simplicity are known merits of the SRM drive.

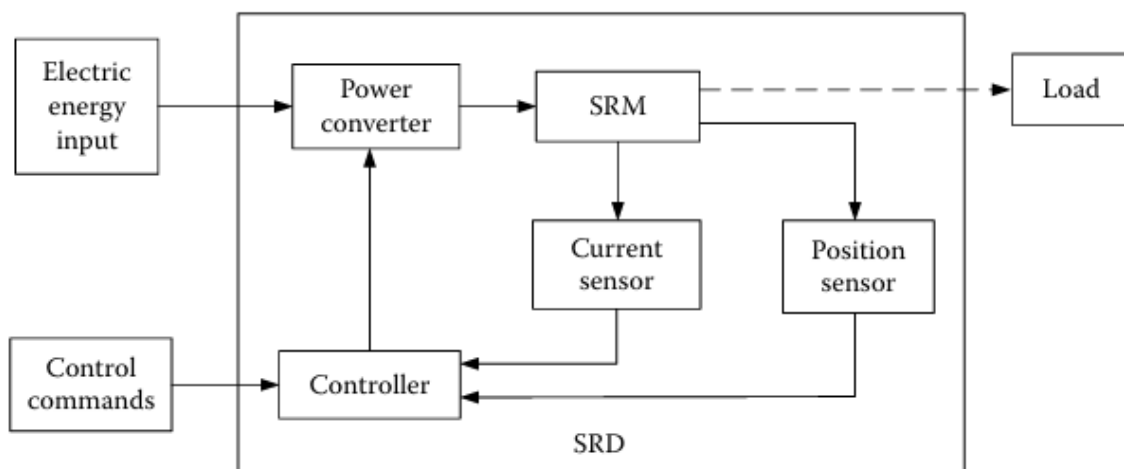


Figure 1: SRM drive system

A conventional SRM drive system consists of the SRM, power inverter, sensors such as voltage, current, and position sensors, and control circuitry such as the DSP controller and its peripherals as shown in Figure 1. Through proper control, high performance can be achieved in the SRM drive system. The SRM drive inverter is connected to a DC power supply, which can be derived from the utility lines through a front-end diode rectifier or from batteries. The phase windings of the SRM are connected to the power inverter, as shown in Figure 2. The control

circuit provides a gating signal to the switches of the inverter according to particular control strategies and the signals from various sensors.

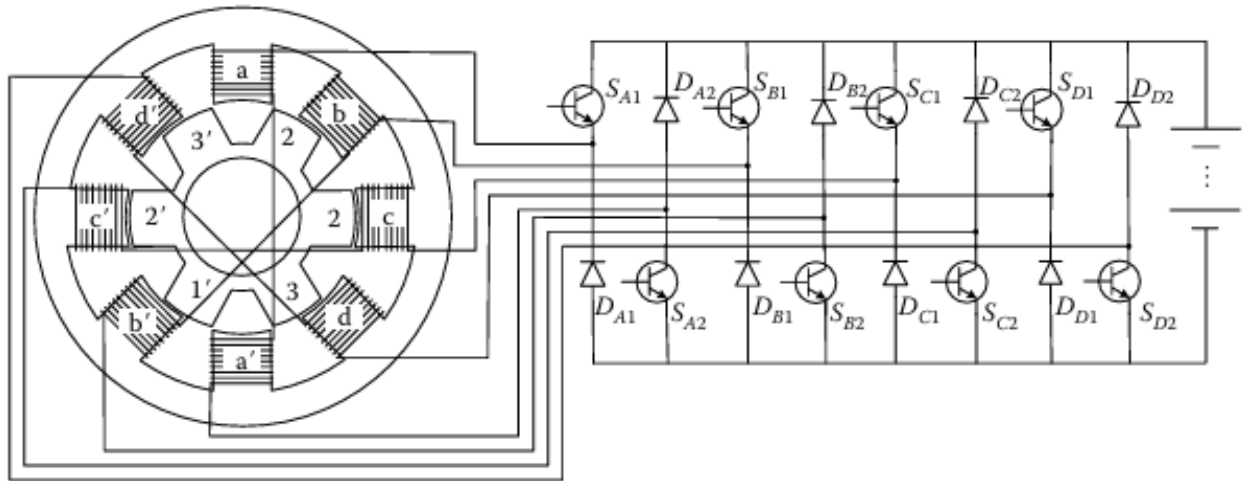


Figure 2: SRM and its power supply

BASIC MAGNETIC STRUCTURE

The SRM has salient poles on both the stator and rotor. It has concentrated windings on the stator and no winding or PM on the rotor. There are several configurations for SRM depending on the number and size of the rotor and stator poles. The configurations of the 8/6 and 6/4 SRMs, which are more common, are shown in Figure 3.

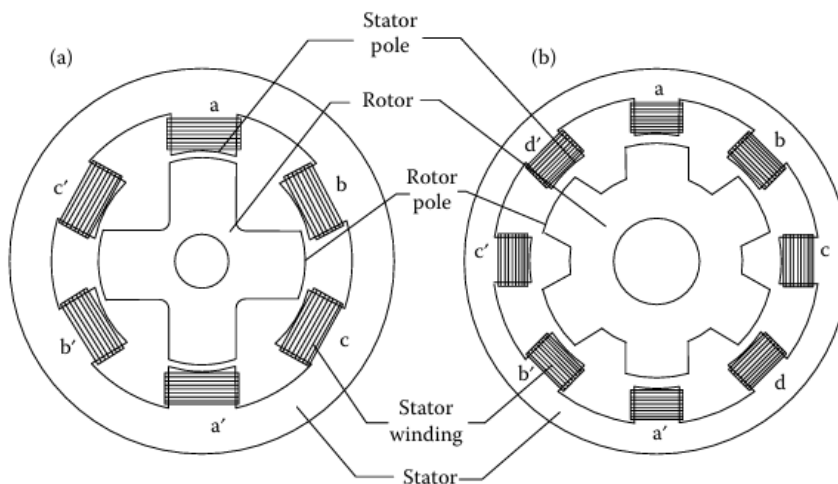


Figure 3: Cross section of common SRM configurations: (a) a 6/4 SRM and (b) an 8/6 SRM

Due to its double saliency structure, the reluctance of the flux path for a phase winding varies with the rotor position. Also, since the SRM is commonly designed for high-degree saturation at

high phase current, the reluctance of the flux path also varies as the phase current. As a result, the stator flux linkage, phase bulk inductance, and phase incremental inductance all vary with the rotor position and phase current.

The phase voltage equation of the SRM (Figure 2) is given by

$$V_j = Ri_j + \frac{d}{dt} \sum_{k=1}^m \lambda_{jk}$$

where m is the total number of phases, V_j is the applied voltage to phase j , i_j is the current in phase j , R is the winding resistance per phase, λ_{jk} is the flux linkage of phase j due to the current of phase k , and t is the time. The phase flux linkage λ_{jk} is given by

$$\lambda_{jk} = L_{jk}(i_k, \theta) i_k$$

where L_{jk} is the mutual inductance between phase k and phase j . The mutual inductance between phases is usually small compared to the bulk inductance and is neglected in equations.

At a fixed phase current, as the rotor moves from the unaligned to the aligned position, the reluctance of the flux path reduces due to the reduction in the air gap. As a result, the phase inductance and flux linkage increases as the rotor moves. At a fixed rotor position, as the phase current increases, the flux path becomes more and more saturated. Hence, the reluctance of the flux path reduces as the phase current increases. As a result, the phase bulk inductance drops with an increase in the phase current. But the phase flux linkage still increases as the phase current increases due to the enhancement in the excitation. The variations of the phase bulk inductance and flux linkage with respect to the phase current and rotor position for an 8/6 SRM are shown in Figures 4 and 5, respectively.

In those figures, $\theta = 30^\circ$ and $\theta = 0^\circ$ represent the unaligned and aligned rotor positions of the referred SRM, respectively.

$$\begin{aligned} V_j &= Ri_j + \frac{d}{dt} \sum_{k=1}^m \lambda_{jk} = Ri_j + \sum_{k=1}^m \left\{ \frac{\partial \lambda_{jk}}{\partial i_k} \frac{di_k}{dt} + \frac{\partial \lambda_{jk}}{\partial \theta} \frac{d\theta}{dt} \right\} \\ &= Ri_j + \sum_{k=1}^m \left\{ \frac{\partial (L_{jk} i_k)}{\partial i_k} \frac{di_k}{dt} + \frac{\partial (L_{jk} i_k)}{\partial \theta} \frac{d\theta}{dt} \right\} \end{aligned}$$

$$= Ri_j + \sum_{k=1}^m \left\{ \left(L_{jk} + i_k \frac{dL_{jk}}{di_k} \right) \frac{di_k}{dt} + i_k \frac{\partial L_{jk}}{\partial \theta} \omega \right\}$$

When only one phase is energized in the operation, Equation can be written as

$$Vi = Ri_j + \left(L_{jj} + i_j \frac{dL_{jj}}{di_j} \right) \frac{di_j}{dt} + i_j \frac{\partial L_{jj}}{\partial \theta} \omega$$

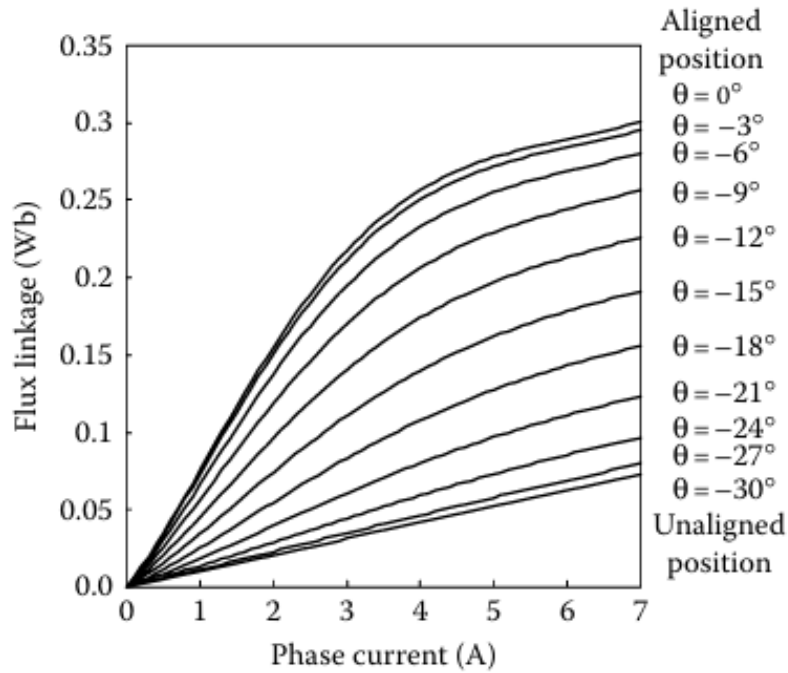


Figure 4: Variation of phase flux linkage with rotor position and phase current

The third term on the right-hand side of Equation is the back EMF. The phase incremental inductance is defined as the derivative of the phase flux linkage against the phase current as

$$\ell_{jj} = \frac{\partial \lambda_{jj}}{i_j} = L_{jj} + i_j \frac{dL_{jj}}{di_j}$$

where $\ell_{jj}(i, \theta)$ and $L_{jj}(i, \theta)$ are phase incremental inductance and bulk inductance, respectively. Figure 4 shows a typical example of flux linkage varying with rotor position, θ , and phase current i of an SRM. Figure 5 shows typical variation of phase bulk inductance with rotor position and phase current.

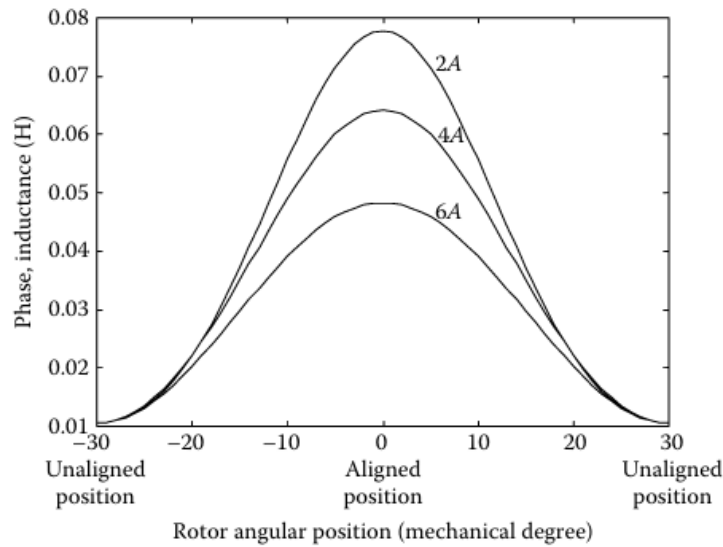


Figure 5: Variation of phase bulk inductance with rotor position and phase current

When the magnetic flux is not saturated, the flux linkage varies linearly with the phase current. The incremental inductance can be viewed as equal to the phase bulk inductance. However, if the machine is saturated at a certain phase current and rotor position, the phase incremental inductance does not equal the phase bulk inductance any more. The variation of the phase incremental inductance with respect to the phase current and rotor position can be derived from the variation of the phase linkage with respect to the phase current and rotor position. The variation of the phase incremental inductance with respect to the phase current and rotor position for an 8/6 SRM is shown in Figure 6.

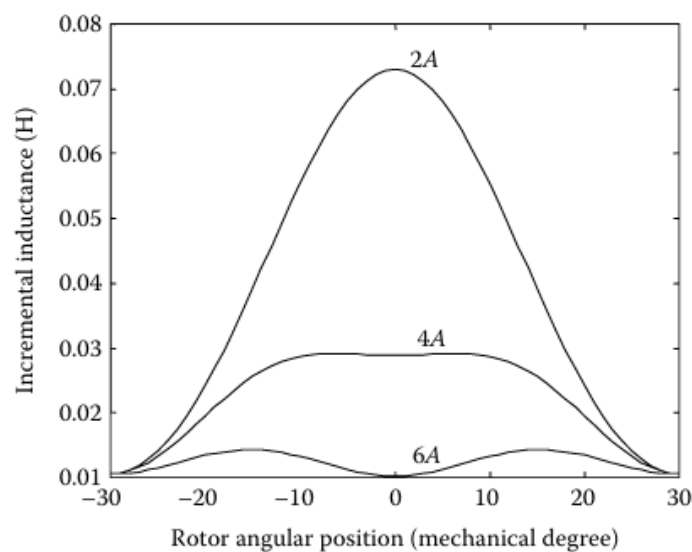


Figure 6: Variation of phase incremental inductance with rotor position and phase current for a typical 8/6 SRM.

TORQUE PRODUCTION

Torque in SRM is produced by the tendency of the rotor to get into alignment with the excited stator poles. The analytical expression of the torque can be derived using the derivative of the coenergy against the rotor position at a given current.

For a phase coil with current i linking a flux λ , the stored field energy W_f and the coenergy W_f' are indicated as shaded regions as shown in Figure 7.

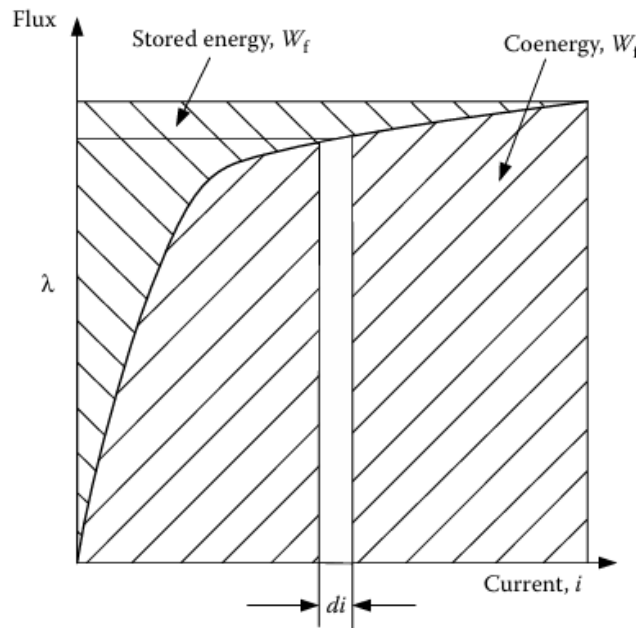


Figure 7: Stored field energy and coenergy

Coenergy can be found from the definite integral:

$$W_f' = \int_0^i \lambda di$$

The torque produced by one phase coil at any rotor position is given by

$$T = \left[\frac{\partial W_f'}{\partial \theta} \right]_{i=\text{constant}}$$

In the case of flux being linear with current, for example, unsaturated field, the magnetization curve in Figure 7 would be a straight line and the coenergy would be equal to the stored field energy. The instantaneous torque can be given as

$$T = \frac{1}{2} i^2 \frac{dL(\theta)}{d\theta}$$

where L is the unsaturated phase bulk inductance.

In the case of saturated phase, the torque cannot be calculated by a simple algebra equation; instead, an integral equation such as

$$T = \int_0^i \frac{\partial L(\theta, i)}{\partial \theta} i di$$

is used

From Equations, it can be seen that in order to produce positive torque (motoring torque) in SRM, the phase has to be excited when the phase bulk inductance increases as the rotor rotates. It can also be observed from Equation that the phase current can be unidirectional for motoring torque production. Hence, low-cost and reliable inverter topology introduced in the later section can be employed for the SRM drive.

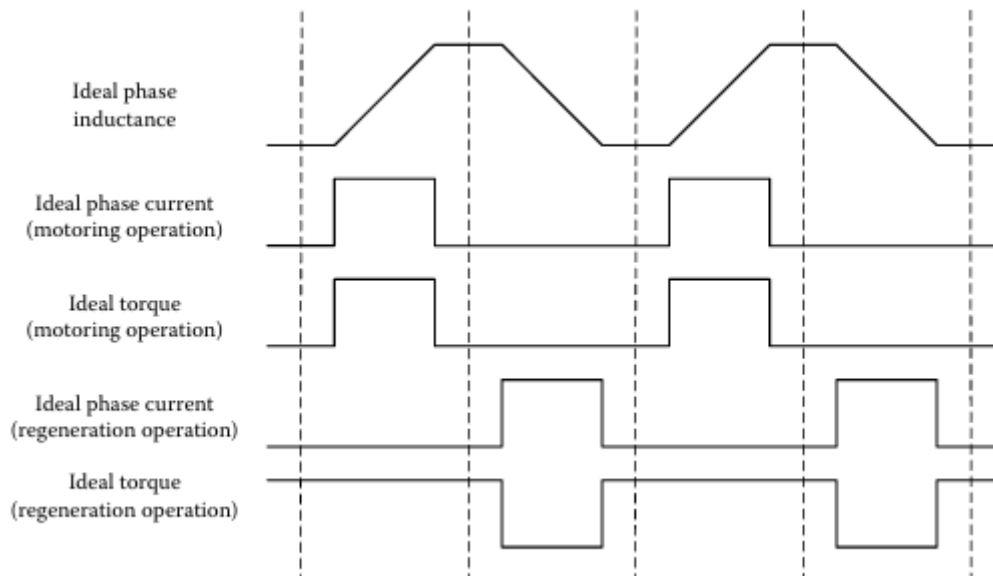


Figure 8: Idealized inductance, current, and torque profiles of the SRM.

Figure 8 shows the ideal phase inductance, current, and torque of the SRM. Positive (motoring) torque is produced if the phase is excited when the phase inductance is increasing as the rotor rotates. Negative torque is generated if the phase is excited when the phase inductance is decreasing as the rotor moves. This implies that the position information is necessary for control of the SRM drive.

The output torque of an SRM is the summation of torque of all the phases:

$$T_m = \sum_{i=1}^N T(i, \theta)$$

Where T_m and N are the output torque and phase number of the motor. The relation between the motor torque and mechanical load is usually given by

$$T_m - T_l = J \frac{d\omega}{dt} + B\omega$$

where J , B , and T_l are the moment of inertia, viscous friction, and load torque, respectively. The relation between position and speed is given by

$$\omega = \frac{d\theta}{dt}$$

SRM DRIVE CONVERTER

It can be seen from Figure 8 that the torque developed by the motor can be controlled by varying the amplitude and the timing of the current pulses in synchronism with the rotor position. In order to control the amplitude and pulse width of the phase current, a certain type of inverter should be used.

The input to the SRM drive is DC voltage, which is usually derived from the utility through a front-end diode rectifier or from batteries. Unlike other AC machines, the currents in SRMs can be unidirectional. Hence, conventional bridge inverters used in AC motor drives are not used in SRM drives. Several configurations have been proposed for an SRM inverter in the literature, some of the most commonly used ones are shown in Figure 9.

The most commonly used inverter uses two switches and two freewheeling diodes per phase and is called the classic converter. The configuration of the classic converter is shown in Figure 9a. The main advantage of the classic converter is the flexibility in control. All the phases can be controlled independently, which is essential for very high-speed operation where there will be considerable overlap between the adjacent phase currents.

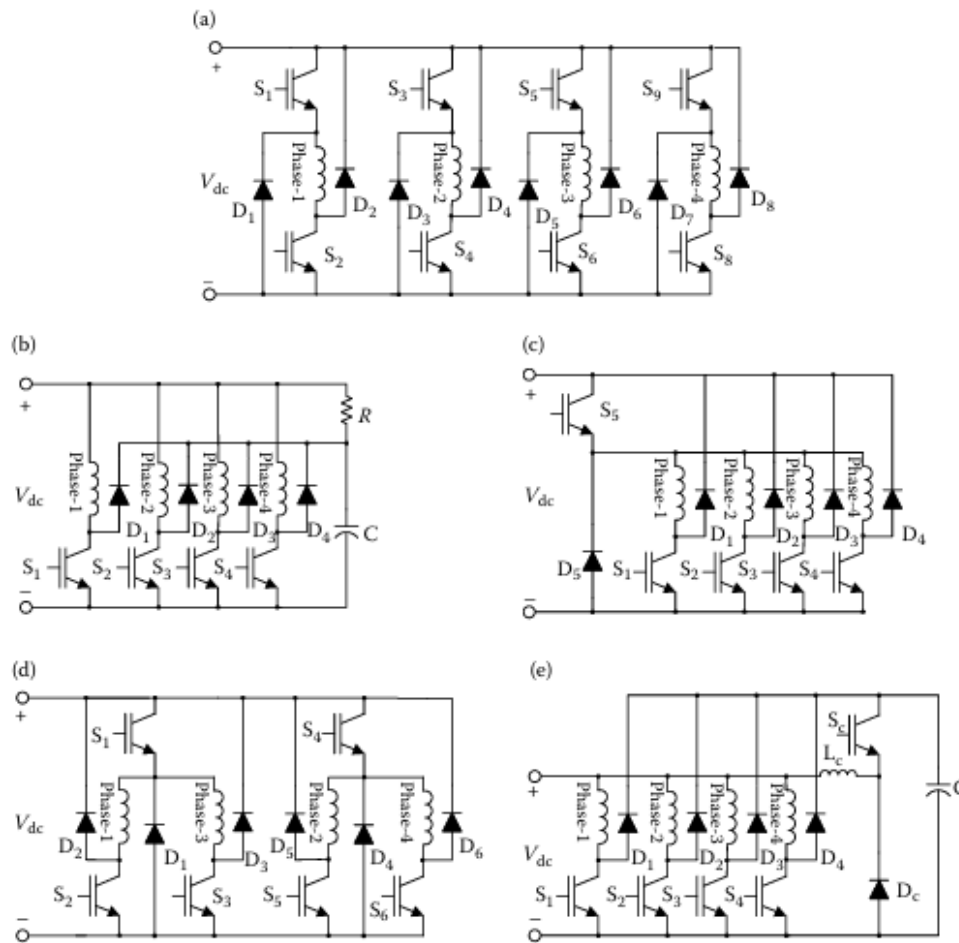


Figure 9: Different inverter topologies for SRM drives: (a) classic half-bridge converter; (b) R-dump; (c) $n + 1$ switches (Miller converter); (d) $1.5n$ switch converter; and (e) C-dump.

The operation of the classic converter is shown in Figure 10 by taking phase-1 as an example. When the two switches S_1 and S_2 are turned on as in Figure 8a, the DC bus voltage, V_{dc} , will be applied to the phase-1 winding. Phase-1 current will increase as it flows through the path consisting of the V_{dc} positive terminal, S_1 , phase-1 winding, S_2 , and the V_{dc} negative terminal.

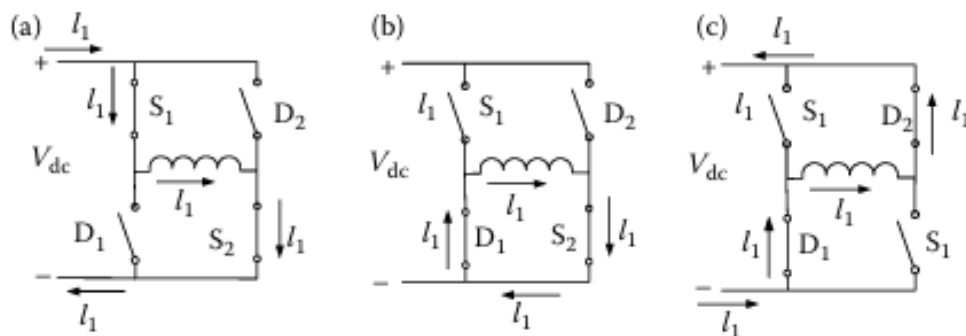


Figure 10: Modes of operation for the classic converter: (a) turn-on mode, (b) zero voltage mode, and (c) turn-off mode.

By turning off S_1 and holding on S_2 (i.e., Figure 6.10b), when the phase is energized, the current freewheels through the S_2 and D_1 . In this mode, phase-1 is not getting or giving energy to the power supply. When S_1 and S_2 are turned off (Figure 8c), the phase-1 current will flow through D_2 , the V_{dc} positive terminal, the V_{dc} negative terminal, D_1 , and phase-1 winding. During this time, the motor phase is subjected to negative DC bus voltage through the freewheeling diodes. The energy trapped in the magnetic circuit is returned to the DC link. The phase current drops due to the negative applied phase voltage. By turning on and off S_1 and S_2 , the phase-1 current can be regulated. The half-bridge converter uses $2n$ switches and $2n$ diodes for an n -phase machine. There are several configurations that use less switches; for example, the R-dump-type inverter (Figure 9b) uses one switch and one diode per phase. This drive is not efficient; during turn-off, the stored energy of the phase is charging capacitor C to the bus voltage and dissipating in resistor R. Also zero voltage mode does not exist in this configuration.

An alternative configuration is $(n + 1)$ switch inverter. In this inverter, all the phases are sharing a switch and diode so that overlapping operation between phases is not possible, which is inevitable in high-speed operation of this motor. This problem has been solved by sharing switches of each couple nonadjacent phases, as shown in Figure 9d. This configuration is limited to an even number of phases of SRM drives.

One of the popular inverter configurations is C-dump (Figure 9e), which has the advantage of less switches and allows independent phase current control. In this configuration, during the turn-off time, the stored magnetic energy is charging capacitor C, and if the voltage of the capacitor reaches a certain value, for example V_c , it is transferred to the supply through the switch S_c . The main disadvantage of this configuration is that the negative voltage across the phase coil is limited to the difference between the voltage across the capacitor V_c and the system power supply voltage.

MODE OF OPERATION

For SRM, there is a speed at which the back EMF is equal to the DC bus voltage. This speed is defined as the base speed. Below the base speed, the back EMF is lower than the DC bus voltage. From Equation $W_f' = \int_0^i \lambda di$, it can be seen that when the converter switches are turned on or off to energize or de-energize the phase, the phase current will rise or drop accordingly. The phase current amplitude can be regulated from 0 to the rated value by turning on or off the

switches. Maximum torque is available in this case when the phase is turned on at an unaligned position and turned off at the aligned position and the phase current is regulated at the rated value by hysteresis or PWM control. The typical waveforms of the phase current, voltage, and flux linkage of the SRM below the base speed are shown in Figure 11.

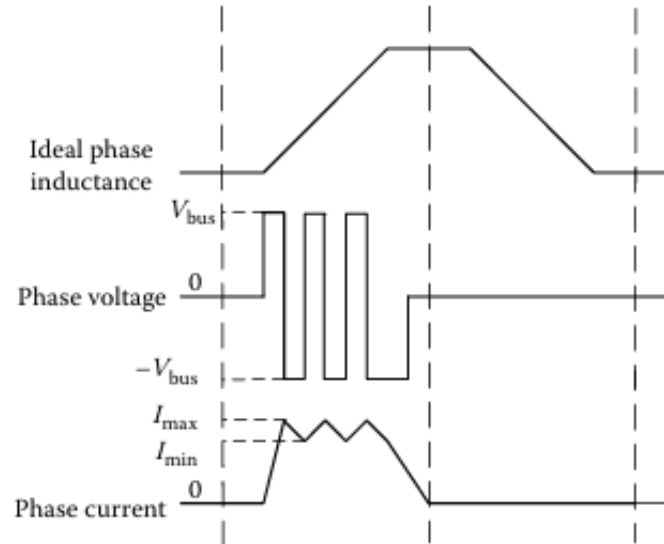


Figure 11: Low-speed (below the base speed) operation of SRM.

Above the base speed, the back EMF is higher than the DC bus voltage. At the rotor position—at which the phase has a positive inductance slope with respect to the rotor position—the phase current may drop even if the switches of the power inverter are turned on. The phase current is limited by the back EMF. In order to build high current and therefore produce high motoring torque in the SRM, the phase is usually excited ahead of the unaligned position, and the turn-on position is gradually advanced as the rotor speed increases. The back EMF increases with the rotor speed. This leads to a decrease in the phase current and hence the torque drops. If the turn-on position is advanced for building as high as possible a current in the SRM phase, the maximum SRM torque almost drops as a linear function of the reciprocal of the rotor speed. The maximum power of the SRM drive is almost constant. The typical waveforms at high-speed operation are shown in Figure 12.

The advancing of the phase turn-on position is limited to the position at which the phase inductance has a negative slope with respect to the rotor position. If the speed of the rotor further increases, no phase advancing is available for building higher current in the phase and the

torque of the SRM will drop significantly. The mode is referred to as the natural mode operation. The torque–speed characteristic of the SRM is shown in Figure 13.

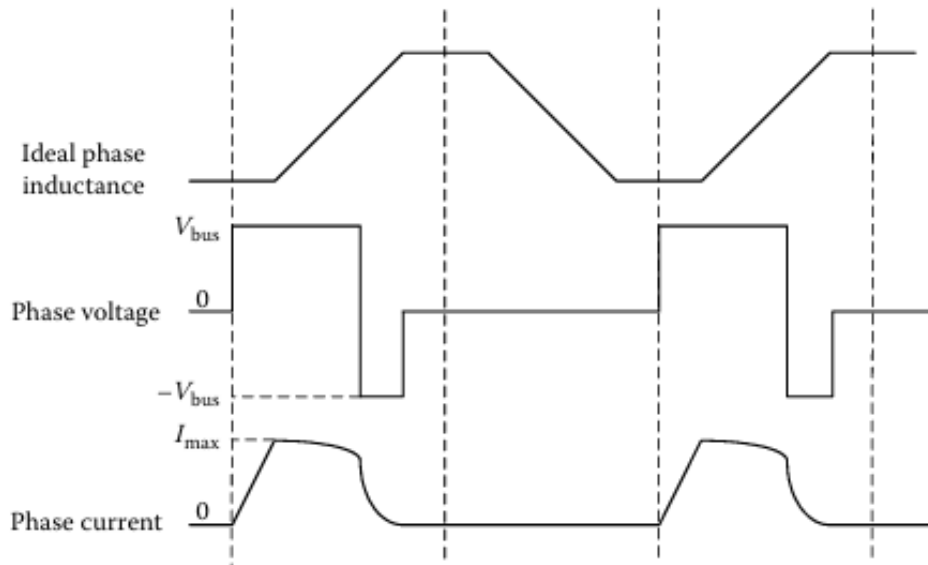


Figure 12: Low -speed (below the base speed) operation of SRM

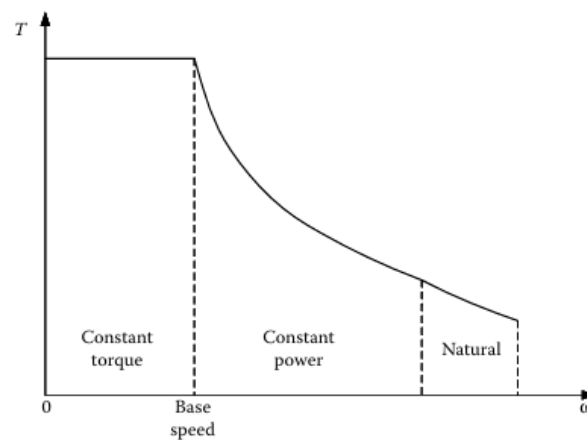


Figure 13: Torque-speed characteristics of SRM

GENERATING MODE OF OPERATION (REGENERATIVE BRAKING)

Torque in the SRMs is created based on the principle of reaching the minimum reluctance for the excited phase. Therefore, if the rotor pole is approaching the excited phase, which means the bulk inductance is increasing, the torque produced is in the direction of the rotor and it is in motoring mode. But if the rotor pole is leaving the stator phase—which means the negative slope of the bulk inductance—the stator tries to keep it in alignment; the torque produced is then in the opposite direction of the movement of the rotor, and the SRM works in the generating mode.

Regenerative braking is an important issue in the propulsion drive of EVs and HEVs. There is a duality in operation of generating and motoring modes and the current waveforms in the generating modes are simply the mirror images of the waveforms in the motoring region around the alignment rotor position. The switched reluctance generator (SRG) is a singly excited machine, so in order to get power from it, it should be excited near the rotor aligned position and then turned off before the unaligned region (Figure 14).

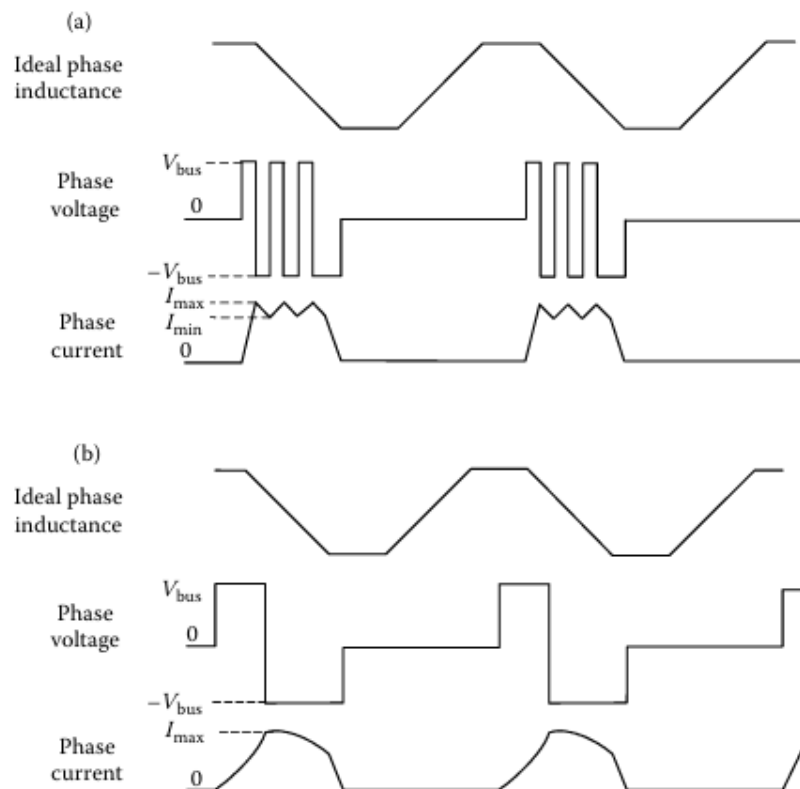


Figure 14: Low and high speed operation in generating mode: (a) low speed operation and (b) high speed operation.

As in motoring operation, current can be controlled by changing the turn-on and turn-off angles and current level while at low speed. Alternatively, at speeds higher than the base speed, only the turn-on and turn-off angles can be used for control.

The driving circuit for SRG is similar to the SRMs; one of the common configurations is shown in Figure 15. When the switches are turned on, the phase gets energy from the supply and capacitor. During the turn-off period, the freewheeling current from the motor charges up the capacitor and delivers energy to the load. Since there is no PM in this motor, during the start-up and initial condition, it needs an external source such as a battery to deliver energy to the phase; after taking transient time, the capacitor is then charged up to the output voltage.

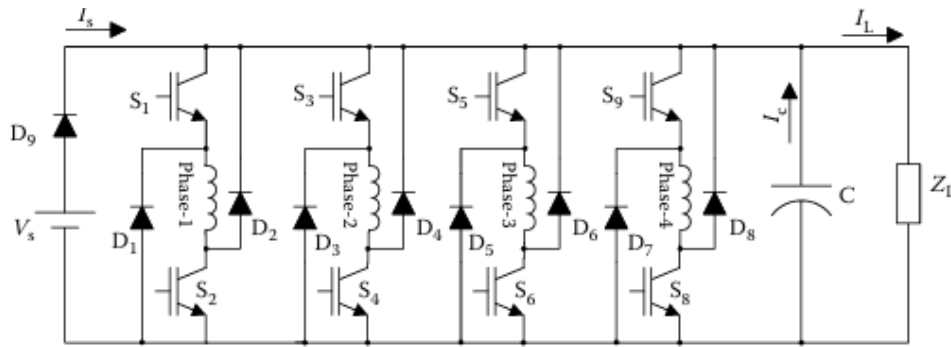


Figure 15: A driving circuit example for the SRG.

Depending on the output voltage during phase-on time, both the capacitor and external source, or just the capacitor, provide the current to the load and the phase coil. The external source can be designed to be charged, or can be disconnected from the system after the system reaches its operating point.

In the generating region, the back EMF is negative so it helps the phase to be charged very fast; then during turn-off the back EMF opposes the negative supply voltage, and it decreases slowly.

$$V_c - e = L \frac{di}{dt} + Ri, e > 0 \text{ (during phase-on period)}$$

$$-V_c - e = L \frac{di}{dt} + Ri, e > 0 \text{ (during phase-off period)}$$

In Equations, V_c is the bus voltage of the inverter or, equivalently, the voltage of the bus capacitor and e is the back EMF voltage. In certain conditions such as high speed and high loads, the back EMF voltage is greater than the bus supply voltage, so the current increases even after turning off the phase. In addition to uncontrollable torque, this necessitates an oversized converter, thereby adding to the cost and overall size of the system. Due to variations in speed of the prime mover, the power electronic converter should be designed for the worst possible case. This will magnify the additional cost and size issues. By properly selecting the turn-off angle, this maximum generating current can be coaxed into the safe region.

Figure 6.69 shows the effect of turn-off angle in the maximum generating current.

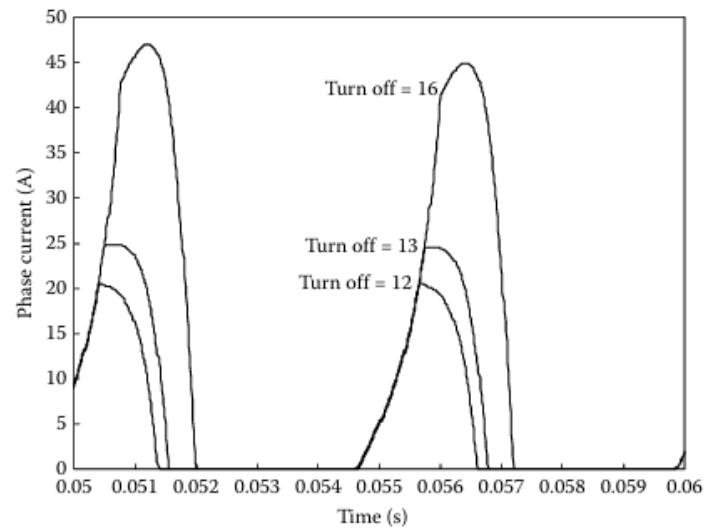


Figure 16: Effect of the turn-off angle in maximum current level in generating mode in an 8/6 switched reluctance machine.